

HSHCS Robotics

Unit 1: Engineering Foundations & FTC Mobility

Two-Week Scope & Sequence • REV V2 • 2026–2027 FTC Preparation

Length	10 × 45-minute classes	Primary Hardware	REV EDU Kit V2
Primary Skill Builders	Fast Lane, Navig8r, Wayfinder	Deferred until V3	Hungry Hungry Robots, Robo-Putt, Hive Hustle
Unit Focus	Build → Drive → Navigate → Program	Culminating Product	Functional drive base + basic autonomous sequence

Unit Rationale

Because the current REV V2 kits do not include the intake/outtake mechanism needed for several 2026–2027 FTC Skill Builders, this unit deliberately prioritizes the skills students can develop well now: drivetrain construction, robot systems, safe wiring, driver control, maneuverability, route planning, consistency, data-based improvement, and introductory autonomous programming. Intake/outtake challenges are postponed until the REV V3 hardware arrives.

Curriculum Alignment

This unit is aligned to the six curriculum strands being used for the HSHCS Robotics Club/Class curriculum. Five strands are directly addressed now; AI & Computer Vision is intentionally deferred to a later unit.

HSHCS Curriculum Strand	Unit 1 Connection	Evidence of Learning
1. Foundations & Safety	Lab routines, team roles, tool/electrical safety, quality control	Safe setup/cleanup, role rotation, build checkpoints
2. Mechanical Design	REV V2 chassis/drivetrain assembly, alignment, wheel and motor systems	Functional, square, free-rolling drivetrain; build notes
3. Electronics & Sensors	Battery, Control Hub, motor connections, encoders as a concept	Correct wiring, component identification, troubleshooting
4. Robot Programming	TeleOp control, sequential logic, timed movement, encoder introduction	Basic autonomous forward/turn sequence
5. AI & Computer Vision	Not a focus in this unit	Deferred to later curriculum
6. FTC Competition Readiness	Fast Lane, Navig8r, driver practice, strategy, scoring, autonomous preparation	Skill Builder performance, data analysis, FTC Skills Passport

Unit Essential Question

How can engineers design, control, test, and program a robot so that it moves accurately, consistently, and strategically?

End-of-Unit Student Outcomes

- Identify major mechanical and electrical systems on an FTC robot.
- Collaboratively assemble and inspect a REV V2 drivetrain.
- Safely connect and test the basic control system.
- Drive forward, reverse, turn, stop, and park with increasing precision.
- Plan and revise a route based on risk, distance, and penalties.
- Collect and interpret performance data across repeated trials.
- Explain the difference between driver-controlled and autonomous operation.
- Create or modify a basic autonomous sequence using forward movement and turning.

Two-Week Scope & Sequence

Day	Lesson / Focus	Learning Target	Key Activities	Skill Builder	Curriculum Alignment	Assessment / Evidence	Materials
1	Robot Systems + Drivetrain Build I	I can identify major robot systems and explain the purpose of a drivetrain.	Robot-system analysis; assign rotating roles; parts inventory; begin chassis assembly.	Preparation	Foundations & Safety; Mechanical Design	Exit ticket; build checkpoint; notebook BUILD entry	REV V2 kits, build guide, trays, hand tools
2	Drivetrain Build II + Quality Control	I can assemble and inspect a drivetrain for alignment, stability, and free movement.	Complete chassis; motor/wheel mounting; frame-square check; wheel free-spin test.	Preparation	Mechanical Design; Foundations & Safety	QC checklist; team reflection; teacher checkpoint	REV V2 kits, straightedge/ruler, hand tools
3	Electronics + Robot Bring-Up	I can identify and safely connect the basic FTC control system.	Component ID; wiring diagram; connect battery/Control Hub/motors; test one motor at a time.	Preparation	Electronics & Sensors; Foundations & Safety	Wiring diagram; component oral check; successful motor test	Control Hub, battery, Driver Hub/gamepad, cables
4	Basic Driver Control Lab	I can control robot movement using smooth, deliberate inputs.	Forward/reverse; precision stop; 90°/180° turns; record overshoot/under shoot.	Fast Lane preparation	FTC Readiness; Robot Programming	Driver-control data sheet; exit ticket	Tape, rulers, robots, timers
5	Fast Lane Practice	I can complete a course while balancing speed and accuracy.	Learn track; rotate driver/coach/referee/timer/score keeper; multiple practice runs.	Fast Lane	FTC Readiness; Foundations & Safety	Practice scores; driver reflection; role accountability	Taped track, timer, score sheets
6	Fast Lane Competition +	I can use performance	3 scored runs per driver;	Fast Lane	FTC Readiness; Data-informed	Official score sheet;	Fast Lane field, timer, score

	Analysis	data to improve driving consistency.	laps/penalties; compare speed vs clean runs; identify improvement.		engineering	improvement statement	sheets
7	Navig8r: Route Planning	I can plan a route that balances distance, risk, and accuracy.	Map course; compare shortest/safest routes; mark turns/risk zones; first test drive.	Navig8r	FTC Readiness; Mechanical/strategic thinking	Annotated route map; prediction vs result	Tape, cups/cones, route maps
8	Navig8r Competition	I can navigate obstacles efficiently while maintaining control.	3 timed runs; obstacle penalties; parking accuracy; strategy revision between runs.	Navig8r	FTC Readiness	Timed/penalty data; strategy reflection	Course obstacles, timer, score sheets
9	Precision Maneuverability Lab	I can perform common FTC maneuvers accurately and repeatedly.	Stations: stop line, 90° alignment, slalom, reverse park, narrow passage; 5-run reliability test.	Bridge to Wayfinder	FTC Readiness; Data analysis	Station passport; reliability %; one evidence-based improvement	Tape, cones/cups, rulers, station cards
10	Wayfinder: Autonomous Foundations	I can break movement into precise programmable steps.	Human-robot activity; forward-stop; forward-turn-stop; timed vs encoder concept; test/debug.	Wayfinder	Robot Programming; Electronics & Sensors; FTC Readiness	Working basic autonomous sequence; debugging log; exit ticket	Robots, programming device, taped mini-course

Daily Instructional Routine

Time	Routine	Purpose
0–5 min	Do Now / retrieval prompt	Activate prior learning and set the engineering problem.
5–12 min	Mini-lesson / demonstration	Teach one focused concept only.
12–35 min	Build / drive / test / program	Maximize hands-on engineering time.
35–40 min	Data + cleanup	Record evidence and restore the lab.
40–45 min	Exit ticket / notebook	Make individual thinking visible.

Rotating Team Roles

Role	Responsibility
Mechanical Engineer	Leads assembly and physical adjustments.
Driver	Operates the robot during assigned trials.
Coach / Strategist	Plans routes and communicates strategy.
Test Engineer	Controls test conditions and repeated trials.
Data Recorder	Records times, penalties, distances, and observations.
Quality-Control Engineer	Checks alignment, hardware, battery, and readiness.
Programmer	Edits and tests autonomous code when programming begins.
Field Manager	Resets course elements and supports safe transitions.

Engineering Notebook Routine

BUILD: What did we assemble or modify?

TEST: What did we test?

DATA: What happened? Include measurements, time, score, or observations.

ANALYSIS: Why do we think it happened?

NEXT: What should we change or test next?

Assessment Plan

Category	Weight	Evidence
Engineering process & documentation	30%	Notebook entries, build checkpoints, data tables, route maps
Collaboration & responsibility	25%	Role completion, safety, equipment care, teamwork
Technical skill development	25%	Skills Passport demonstrations, driving, wiring, autonomous tasks
Reflection & data analysis	20%	Exit tickets, improvement analysis,

		reliability calculations
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FTC Foundations Skills Passport

Domain	Student Demonstrations
Hardware	☐ Identify drivetrain ☐ Identify motors ☐ Identify Control Hub ☐ Identify battery ☐ Explain motor wiring
Driver Control	☐ Drive straight ☐ Reverse accurately ☐ 90° turn ☐ Precision stop ☐ Slalom ☐ Reverse park
Strategy	☐ Plan a route ☐ Identify risk ☐ Compare routes ☐ Revise strategy from evidence
Engineering	☐ Build participation ☐ Record data ☐ Diagnose a problem ☐ Make evidence-based improvement
Programming	☐ Explain autonomous ☐ Program forward ☐ Program stop ☐ Program turn ☐ Explain timed vs encoder movement

Deferred Skill Builders and Phase 2

These challenges are intentionally postponed until the REV V3 intake/outtake hardware is available:

- Hungry Hungry Robots — intake design, acquisition, and reliability.
- Robo-Putt — controlled outtake, aim, power, and repeatability.
- Hive Hustle — integrated autonomous navigation + intake + delivery cycles.

When V3 arrives, students should compare the new hardware to the V2 platform, identify what changed, and explain why the new mechanism enables tasks that were not practical during Unit 1.

Teacher Preparation Checklist

- ☐ Inventory REV V2 kits and verify batteries/chargers.
- ☐ Print or distribute drivetrain build instructions.
- ☐ Prepare labeled team hardware trays.
- ☐ Tape a beginner Fast Lane track approximately 3 ft wide.
- ☐ Prepare a Navig8r field with 4 checkpoints and 5 obstacles for the first run.
- ☐ Create Day 9 maneuverability stations.
- ☐ Confirm Driver Hub/gamepad pairing and programming access before Day 10.
- ☐ Print team role cards, engineering notebook sheet, score sheets, and Skills Passport.

Source Basis

This unit uses the 2026–2027 FIRST Tech Challenge Skill Builders supplied for Fast Lane, Navig8r, and Wayfinder as the primary challenge framework. Fast Lane develops controlled driving and consistency; Navig8r develops route planning, precision, and risk/reward decisions; Wayfinder develops sequential autonomous logic, timed versus encoder-based movement, testing, debugging, and iteration. The unit also preserves the broader HSHCS curriculum strands already established for the robotics class/club and intentionally defers intake/outtake-centered challenges until appropriate hardware is available.